

MacRay-ToF-WXx Wi-Fi Time-of-Flight Sensor Datasheet

Product Overview

MacRay-ToF-WXx is a high-precision **Wi-Fi-based Time-of-Flight (ToF) sensor** designed for **indoor people counting, occupancy analytics, and zone monitoring applications** within Wi-Fi-enabled environments. Using short-range 3D depth sensing with millimeter-level accuracy, it enables real-time space utilization insights, facility automation, and safety monitoring without relying on cameras or capturing personal identity data.

The sensor uses **advanced ToF distance mapping and motion detection algorithms** to detect human presence, count bidirectional movement across defined boundaries, monitor zone activity, and generate accurate occupancy metrics. Its privacy-first design ensures compliance with workplace and public-space privacy requirements.

MacRay-ToF-WXx features a **power-operated design with inbuilt backup battery**, ensuring uninterrupted operation during short power interruptions. It communicates directly over **standard Wi-Fi networks (2.4 GHz / 5 GHz)** to **on-premise servers, building management systems, or cloud platforms** using industry-standard IP protocols such as **MQTT, HTTP, and HTTPS**, without the need for gateways or long-range wireless infrastructure.

With support for **entry/exit counting, room occupancy limits, restricted-zone alerts, and automation triggers**, MacRay-ToF-WXx transforms physical movement data into actionable insights for **smart buildings, offices, hospitals, retail spaces, industrial facilities, and campuses**, enabling improved space efficiency, energy optimization, and operational safety.





Key Features

People Counting & Occupancy Detection

Accurate bidirectional people counting and real-time occupancy analytics using privacy-safe ToF sensing.

Privacy-First Design

No images, video, or personally identifiable data captured, making it ideal for sensitive indoor environments.

Real-Time Events & Automation Triggers

Supports occupancy limits, restricted-zone alerts, and automation triggers for HVAC, lighting, and access control.

Compact Ceiling & Wall Mount Design

Optimized form factor for doors, corridors, rooms, and zone-focused indoor installations.

Secure Wireless Communication

WPA2 / WPA3 Wi-Fi security with TLS-encrypted data transmission.

Time-of-Flight 3D Depth Sensing

Millimeter-level short-range distance and depth measurement for precise zone monitoring and entry/exit detection.

Wi-Fi Based Connectivity

Direct communication over standard Wi-Fi networks (2.4 GHz / 5 GHz) without gateways or long-range infrastructure.

Power-Operated with Backup Battery

Continuous operation with inbuilt backup battery support during short power interruptions.

Standard IP Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless integration with BMS, dashboards, and cloud platforms.

Local Configuration via BLE

Fast provisioning, diagnostics, and configuration using the Macman Maya mobile application.

Suited Applications

People Counting at Entry & Exit Points

Accurate bidirectional people counting at doors, corridors, turnstiles, and access-controlled entry points using privacy-safe ToF sensing.

Smart Buildings & Facility Management

Occupancy-driven automation for HVAC, lighting, and ventilation systems within Wi-Fi-enabled buildings and campuses.

Hospitals & Healthcare Facilities

Privacy-safe monitoring of patient flow, waiting areas, and restricted zones without using cameras.

Warehouses & Industrial Indoor Areas

Tracks workforce movement and zone activity inside warehouses and factories for safety compliance and operational visibility.

Room & Zone Occupancy Monitoring

Real-time occupancy tracking for offices, meeting rooms, classrooms, hospital wards, and conference areas to optimize space usage.

Retail Footfall & Queue Analytics

Customer movement analysis, queue length monitoring, and zone engagement insights for retail stores and shopping spaces.

Restricted & Secure Area Monitoring

Detects unauthorized entry into server rooms, labs, storage areas, and sensitive operational zones.

Smart Campuses & Public Facilities

Occupancy and movement analytics for airports, stations, educational campuses, and large public buildings.

Technical Specifications

General Specifications

Part Order Number	MacRay-ToF-WXx
Product Brand	Macnman Technologies Pvt Ltd

Time-of-Flight Measurement

Distance Range	Long Range Mode: 4 cm to 400 cm * Short Range Mode: 4 cm to 130 cm *
Sensors Grid Options	2 X 2 4 X 4 8 X 8
Accuracy	±3% typical * < ±5 mm (Indoor lighting conditions)
Repeatability	< ±10 mm (under strong ambient light)
Resolution	Up to 1 mm

Measurement Rate	Up to 60 Hz (adjustable)
Field of View (FoV)	Programmable, up to 26° ~ 60°
Ambient Light Immunity	Up to 100k Lux (sunlight)
Wavelength	940 nm (IR, eye-safe & invisible)
Response Time	< 25 ms (typical)
Target Reflectivity Tolerance	Reliable performance on dark, light, & low-reflective surfaces

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacRay-ToF-WXx Variants

VARIANT NAME	SENSOR GRID PATTERN	COVERAGE	RECOMMENDED USE
MacRay-ToF-LX2-PO	2 × 2 (4-point ToF)	Narrow zone detection	Doorway people counting / Access control
MacRay-ToF-LX4-PO	4 × 4 (16-point ToF)	Medium area coverage	Room occupancy analytics / Facility automation
MacRay-ToF-LX8-PO	8 × 8 (64-point ToF)	Wider coverage with higher precision	Zone activity mapping / Retail & smart workspace analytics

What's in the Box?

Make sure box contains following items before installations



MacRay



**Mounting
Clip***



**User manual &
Warranty Card**



**IP
Extender***



**Mounting
Screws**

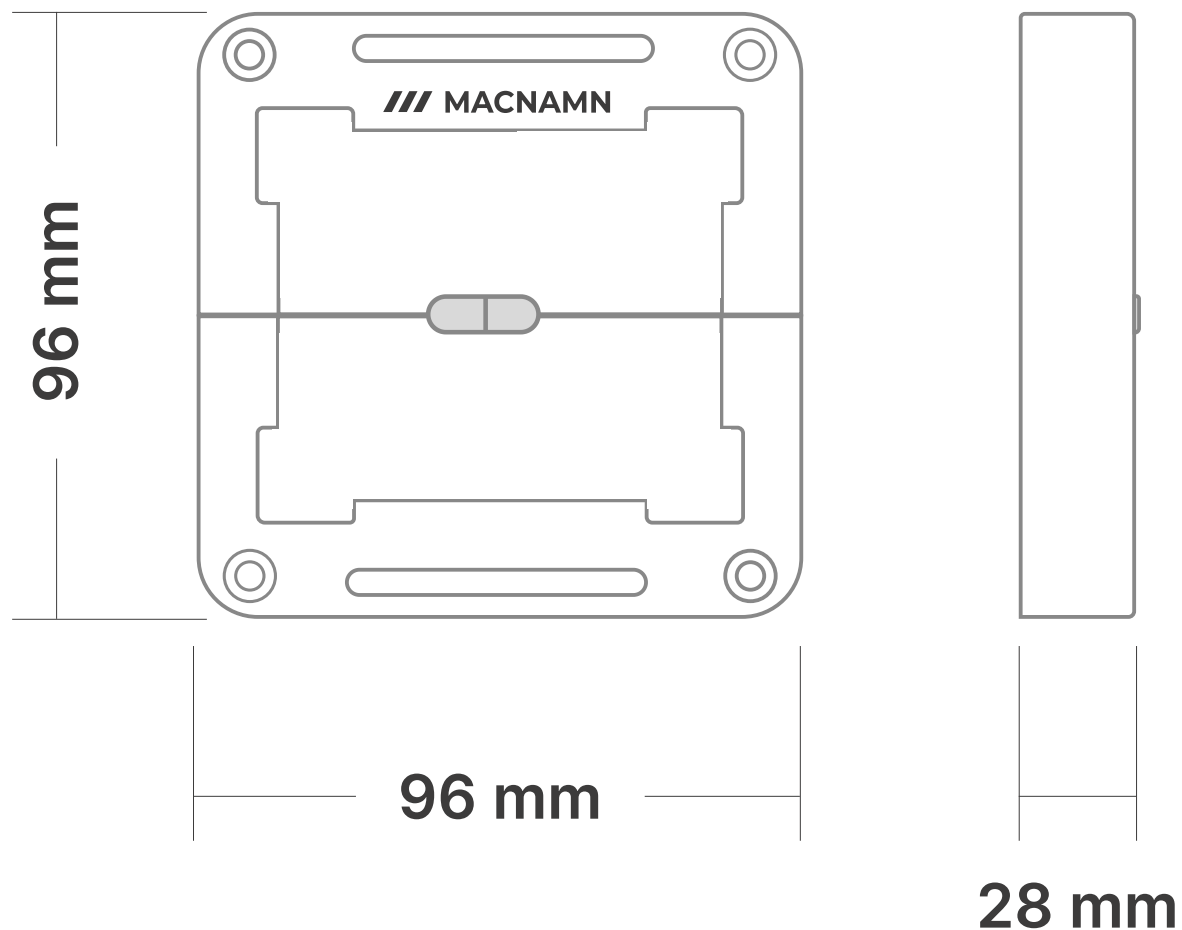


**Power
Adapters***

i Note

Accessories must be purchased separately & may not be included in the box.

MacRay-ToF-WXx Dimensions



Frequently Asked Questions

What is the MacRay-ToF-WXx used for?

MacRay-ToF-WXx is a high-precision Wi-Fi Time-of-Flight sensor for indoor people counting, occupancy analytics and zone monitoring. Typical applications include People Counting at Entry & Exit Points, Room & Zone Occupancy Monitoring, Smart Buildings & Facility Management, Retail Footfall & Queue Analytics, Hospitals & Healthcare Facilities and Restricted & Secure Area Monitoring.

What does the MacRay-ToF-WXx measure, and how accurately?

Measurement specifications of the MacRay-ToF-WXx:

- Time-of-Flight: range Long Range Mode: 4 cm to 400 cm *, Short Range Mode: 4 cm to 130 cm *, accuracy $\pm 3\%$ typical *, $< \pm 5$ mm (Indoor lighting conditions), resolution Up to 1 mm.

How does the MacRay-ToF-WXx connect and send data over Wi-Fi?

The MacRay-ToF-WXx connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacRay-ToF-WXx powered?

The MacRay-ToF-WXx is powered from 9-36 V DC (External Supply).

What is the IP rating and operating temperature of the MacRay-ToF-WXx?

The MacRay-ToF-WXx has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to $+80^{\circ}\text{C}$. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacRay-ToF-WXx configured?

The MacRay-ToF-WXx is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacRay-ToF-WXx work with?

The MacRay-ToF-WXx can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacRay LToF LoRaWAN Time-of-Flight sensor](#), [MacRay-WU4 Wi-Fi ultrasonic level sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)